PRODUCT AND QUOTIENT RULES

BLAKE FARMAN

Lafayette College

Name: Solutions

In each of the problems, use the

Product Rule. If f and g are differentiable functions, then

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(f(x)g(x)\big) = f'(x)g(x) + f(x)g'(x)$$

and

Quotient Rule. If f and g are differentiable functions, then

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$$

to compute the derivative. Use proper notation and simplify your final answers. In some cases it might be advantageous to simplify/rewrite first. **Do not use rules found in later sections.**

1. $f(x) = \left(1 + \sqrt{x}\right) x^3$

$$f'(x) = \frac{d}{dx}\left(x^3 + x^{7/2}\right)$$

$$= \boxed{3x^2 + \frac{7}{2}x^{5/2}}$$

2. $g(t) = \left(\frac{2}{t} + t^5\right)(t^3 + 1)$

$$g'(t) = (-2t^{-2} + 5t^4)(t^3+1) + (2/t + t^5)(3t^2)$$

$$= -2t - 2t^{-2} + 5t^7 + 5t^4 + 6t + 3t^7$$

$$= \boxed{8t^7 + 5t^4 + 4t - 2t^{-2}}$$

3. $h(y) = \frac{1}{y^3 + 2y + 1}$

$$h'(y) = \frac{0(y^3+2y+1) - (1)(3y^2+2)}{(y^3+2y+1)^2}$$

$$= \boxed{\frac{-(3y^2+2)}{(y^3+2y+1)^2}}$$

4. $y = \frac{1}{x + \sqrt{x}}$

$$y' = \frac{0 - (1 + \frac{1}{2\sqrt{x}})}{(x+\sqrt{x})^2} = \left(-\frac{2\sqrt{x}}{2\sqrt{x}} - \frac{1}{2\sqrt{x}}\right) \frac{1}{(x+\sqrt{x})^2}$$

$$= \boxed{\frac{-(1+2\sqrt{x})}{2\sqrt{x}(x+\sqrt{x})^2}}$$

5. $g(z) = \dfrac{z^2+1}{z^3-5}$

$$g'(z) = \frac{(2z)(z^3-5) - (z^2+1)(3z^2)}{(z^3-5)^2}$$

$$= \frac{2z^4 - 10z - 3z^4 - 3z^2}{(z^3-5)^2} = \boxed{\frac{-z^4-3z^2-10z}{(z^3-5)^2}}$$

6. $y = \dfrac{\sqrt{x}}{x^3+1}$

$$y' = \frac{\frac{1}{2\sqrt{x}}(x^3+1) - \sqrt{x}(3x^2)}{(x^3+1)^2} = \frac{x^3+1 - x(3x^2)}{\sqrt{x}} \quad \frac{1}{(x^3+1)^2}$$

$$= \boxed{\frac{1-2x^3}{\sqrt{x}(x^3+1)^2}}$$

7. $z = \dfrac{t^2}{(t-4)(2-t^3)}$ $\qquad (t-4)(2-t^3) = -t^4+4t^3+2t-8$

$$z' = \frac{2t(-t^4+4t^3+2t-8) - t^2(-4t^3+12t^2+2)}{(t-4)^2(2-t^3)^2}$$

$$= \frac{-2t^5+8t^4+4t^2-16t+4t^5-12t^4-2t^2}{(t-4)^2(2-t^3)^2}$$

$$= \boxed{\frac{2t^5-4t^4+2t^2-16t}{(t-4)^2(2-t^3)^2}}$$

8. $h(x) = \dfrac{(x^3+1)\sqrt{x}}{x^2}$

$$h'(x) = \frac{d}{dx}\left(\frac{x^{7/2}+x^{1/2}}{x^{4/2}}\right) = \frac{d}{dx}\left(x^{3/2}+x^{-3/2}\right)$$

$$= \boxed{\tfrac{3}{2}x^{1/2} - \tfrac{3}{2}x^{-1/2}}$$

9. Let $f(x) = g(x)h(x)$, $g(10) = -4$, $h(10) = 560$, $g'(10) = 0$, $h'(10) = 4$. Find $f'(10)$.

$$f'(10) = g'(10)h(10) + g(10)h'(10)$$

$$= 0 - 4(4) = \boxed{-16}$$

10. Let $z(-3) = 6$, $z'(-3) = 15$, and $y(x) = \dfrac{z(x)}{1+x^2}$. Find $y'(-3)$.

$$y'(-3) = \frac{z'(-3)\left(1+(-3)^2\right) - z(-3)\left(2(-3)\right)}{\left(1+(-3)^2\right)^2}$$

$$= \frac{15(10) - 6(-6)}{100} = \frac{150+36}{100} = \frac{186}{100} = \boxed{\frac{93}{50}}$$

Compute the following derivatives using

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) = \cos(x) \qquad \text{and} \qquad \frac{\mathrm{d}}{\mathrm{d}x}\cos(x) = -\sin(x)$$

and the trigonometric identities

- $\tan(x) = \dfrac{\sin(x)}{\cos(x)}$
- $\sec(x) = \dfrac{1}{\cos(x)}$
- $\cot(x) = \dfrac{1}{\tan(x)} = \dfrac{\cos(x)}{\sin(x)}$
- $\csc(x) = \dfrac{1}{\sin(x)}$

11. $\dfrac{\mathrm{d}}{\mathrm{d}x}\tan(x)$

$$= \frac{\cos(x)\cos(x) - \sin(x)(-\sin(x))}{\cos^2(x)}$$

$$= \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)}$$

$$= \frac{1}{\cos^2(x)} = \boxed{\sec^2(x)}$$

12. $\dfrac{\mathrm{d}}{\mathrm{d}x}\cot(x) = \dfrac{(-\sin(x))\sin(x) - \cos(x)(\cos(x))}{\sin^2(x)}$

$$= \frac{-(\sin^2(x) + \cos^2(x))}{\sin^2(x)} = \boxed{-\csc^2(x)}$$

13. $\frac{\mathrm{d}}{\mathrm{d}x}\sec(x)$

$$= \frac{0-(-\sin(x))}{\cos^2(x)} = \frac{\sin(x)}{\cos(x)}\left(\frac{1}{\cos(x)}\right)$$

$$= \boxed{\tan(x)\sec(x)}$$

14. $\frac{\mathrm{d}}{\mathrm{d}x}\csc(x) = \frac{0-\cos(x)}{\sin^2(x)} = -\frac{\cos(x)}{\sin(x)}\frac{1}{\sin(x)}$

$$= \boxed{-\cot(x)\csc(x)}$$